Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Roll No.:XXXXXX youremail@email.com
Your Course officialemail@iitp.ac.in
Mathematics and Computing Github | Website
Indian Institute Of Technology, Patna linkedin.com/in/linkedinuser

1 Education

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Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (MnC)	Indian Institute of Technology, Patna	8.19 (Current)	2021-Present
B.Tech. (CSE)	Institute of Engineering and Management, Kolkata	8.21	2015-2019
Senior Secondary	CBSE Board	88.0%	2014
Secondary	CBSE Board	85.5%	2012

2 Experience

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Tata Consultancy Services *Aug 2019 - Nov. 2020*
Assistant System Engineer Kolkata

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- Managed ETL process for reporting purposes in a team of 4, improved PL/SQL packages for faster and consistent data flow. Reduced month-end activity time by 10hrs, created Alteryx Workflows for Data analysis and validations.
- Developed a plugin that simplified the debugging process by differentiating log messages with a different colour that helped the support team while investigating incidents.

Thinking Stack Inc. *Dec. 2018 - Jan. 2019*
Full Stack Developer Intern Bengaluru, India

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- Developed web app using ReactJS and deployed the app to AWS EC2 using Docker.
- Made website mobile-friendly and increased Google web dev score by 50 points.

3 Projects

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Project Tracker *Feb. 2022*
Developed progress tracker for clients to track payments, progress, meetings.

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- **Tools & technologies used**: NextJs/ReactJS, Strapi/Node, Tailwind/CSS

What To Watch Next *Dec. 2021 - Jan. 2022*
Content based Movie Recommendation system that recommends 5 other movie based on a given movie. Live Url | Github

Kisan Mitra(SIH 2019) *Feb. 2019 - Mar. 2019*
Web Portal for Ministry of Agriculture to view the predicted crop yield data in geo-refrenced map. Github

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- **Tools & technologies used**: QGIS, Java, HTML, CSS, JS, ChartistJS
- Created APIs to fetch crop production data and displayed them into Geo-refrenced Map.

4 Key Courses Taken

[leftmargin=*,itemsep=1.7mm, rightmargin=2ex] Data Structure,Algorithms,DBMS,Machine Learning, Artificial Intelligence, Web Services, Operating System

5 Technical Skills

[leftmargin=*,itemsep=1.7mm, rightmargin=2ex] **Programming:** C/C++,Python,R,JavaScript,SQL **Tools & OS:** Git, Jupyter Notebook,Google Colab, Linux, Windows **Libraries/Frameworks:** Pandas, Numpy, scikit-learn, JQuery **Web Skills:** HTML/CSS/UI/UX

6 Positions of Responsibility

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Assistant Head Coordinator, Training & Placement Cell, IIT Patna *Present*

General Secretary, Nexus Coding Club, IEM Kolkata *Apr. 2018 - Apr. 2019*

7 Achievements

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Secured 1st Position Smart India Hackathon, MHRD, Government of India. Won prize of 1Lakh INR. *2019*

Winner Web Designing Competition(URECKON'18) organized by University of Engineering & Management, ISRO *2018*

Github Arctic Code Vault Contributor Contributed code to several repositories in the 2020 GitHub Archive Program *2020*

8 Certifications

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Datacamp Certification on Introduction to R

NPTEL Certification on Database Management System

Internshala Certification on Web Developmet